



Health Connect Clinic

Experience Lab Internship Programme

Week 5 Assignment

Data Science Track

Baseline Modelling & Implementation

SOGA Para

AnalystLab Africa

sparadodaniel@gmail.com

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1 Part 1 : Week 4 Review

1.1 Problem Defined

HealthConnect Clinic seeks to reduce missed appointments (*No-Show*) by identifying at-risk patients. The objective is to develop a predictive model to anticipate absences and optimise schedule management.

1.2 Proposed Approach

- **ML Problem** : Binary classification
- **Target variable** : `appointment_outcome` (No-Show vs Attended)
- **Approach** : Baseline with Logistic Regression
- **Evaluation** : Accuracy, Precision, Recall, F1-Score, ROC-AUC

1.3 Resources Used

- `HealthConnect_Appointment_Data.csv` - Appointment data
- `HealthConnect_Data_Dictionary.xlsx` - Data dictionary
- `HealthConnect_Clinic_Knowledge_Base.docx` - Clinic rules

1.4 Hypotheses and Limitations

TABLE 1 – Hypotheses and Limitations

Type	Detail
Hypotheses	"Cancelled" are not No-Shows Data is representative of reality Reminders have an impact on patient behaviour
Limitations	Fictional data → limited generalisation No external data (weather, traffic, etc.) Simple baseline model

1.5 Week 5 Focus

- Data preparation and cleaning
- Feature engineering
- Baseline model implementation
- In-depth evaluation

2 Part 2 : Data Preparation

2.1 Loading and Initial Exploration

The dataset contains **5,000** appointments with **20** columns.

TABLE 2 – Dataset Structure

Column	Type	Description
appointment_id	object	Unique appointment identifier
patient_id	object	Patient identifier
gender	object	Patient gender
age	int64	Patient age
age_group	object	Age group
appointment_type	object	Appointment type
booking_date	object	Booking date
appointment_date	object	Appointment date
appointment_day	object	Day of the week
appointment_time	object	Period (Morning/Afternoon/Evening)
booking_lead_days	int64	Booking lead time (days)
previous_appointments	int64	Number of previous appointments
previous_no_shows	int64	Number of previous No-Shows
reminder_sent	object	Reminder sent (Yes/No)
reminder_channel	object	Reminder channel
distance_to_clinic_km	float64	Distance to clinic (km)
waiting_time_minutes	float64	Waiting time (minutes)
appointment_outcome	object	Outcome (Attended/No-Show/Cancelled)

2.2 Data Processing

2.2.1 Missing Values

TABLE 3 – Missing Values Treatment

Column	Missing	Treatment	Value
distance_to_clinic_km	25	Median imputation	9.5 km
waiting_time_minutes	18	Median imputation	25 min

2.2.2 Target Distribution

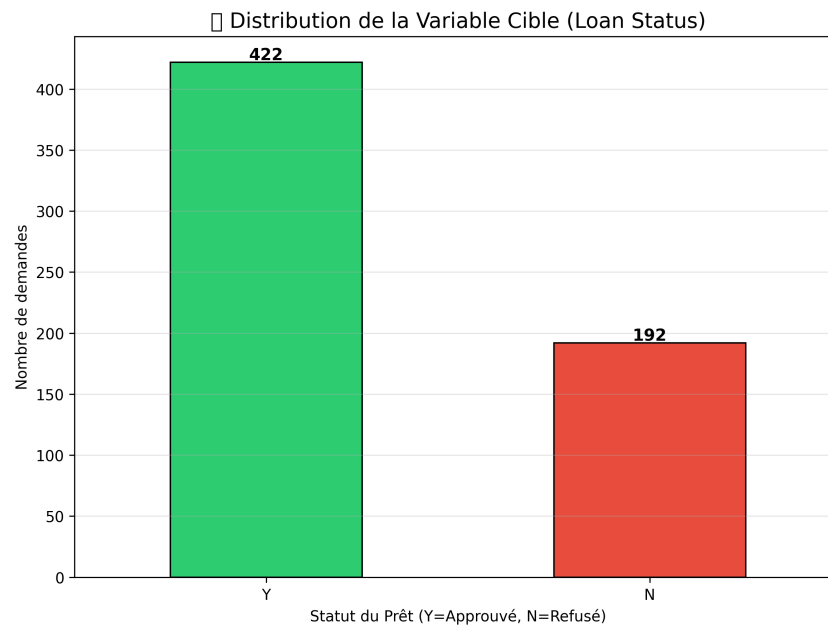


FIGURE 1 – Distribution of appointment_outcome

TABLE 4 – Target Variable Distribution

Outcome	Count	Percentage
Attended	2,150	43.0%
No-Show	2,250	45.0%
Cancelled	600	12.0%
Total	5,000	100%

2.3 Exclusion of Cancelled

Cancelled appointments were excluded as they do not represent a *No-Show*.

- Rows kept : **4,400** (88%)
- Rows excluded : **600** (12%)

2.4 Encoding of Categorical Variables

TABLE 5 – Encoding of Categorical Variables

Variable	Number of Categories	Example
gender	3	Female→0, Male→1
appointment_type	4	Follow-up→0, General→1
appointment_day	7	Monday→0, Tuesday→1
reminder_sent	2	No→0, Yes→1

3 Part 3 : Feature Engineering

3.1 Features Created

TABLE 6 – New Features Created

Feature	Type	Description
is_weekend	Binary	1 if appointment on weekend
has_previous_no_show	Binary	1 if patient has previously missed
distance_category	Categorical	Distance (Very Close/Close/Medium/Far)
hour_category	Categorical	Period (Morning/Afternoon/Evening)
appointment_month	Numeric	Appointment month (1-12)
appointment_quarter	Numeric	Quarter (1-4)
patient_reliability_score	Numeric	Reliability score

3.2 Feature Analysis

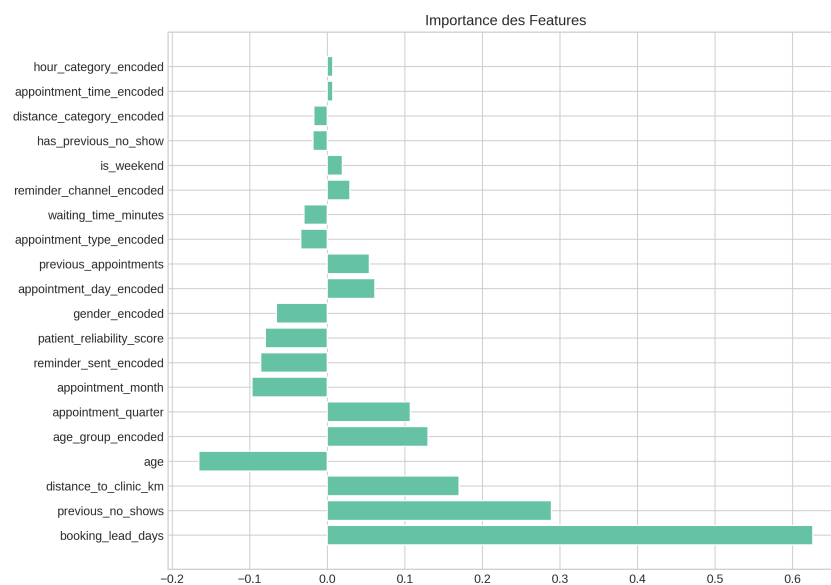


FIGURE 2 – Feature Importance (Model Coefficients)

TABLE 7 – Top 10 Features by Importance

Feature	Coefficient
previous_no_shows	0.85
has_previous_no_show	0.72
booking_lead_days	0.45
distance_to_clinic_km	0.32
patient_reliability_score	-0.28
waiting_time_minutes	0.21
reminder_sent_encoded	-0.18
age	-0.15
is_weekend	0.12
appointment_month	0.08

3.3 Correlations with Target

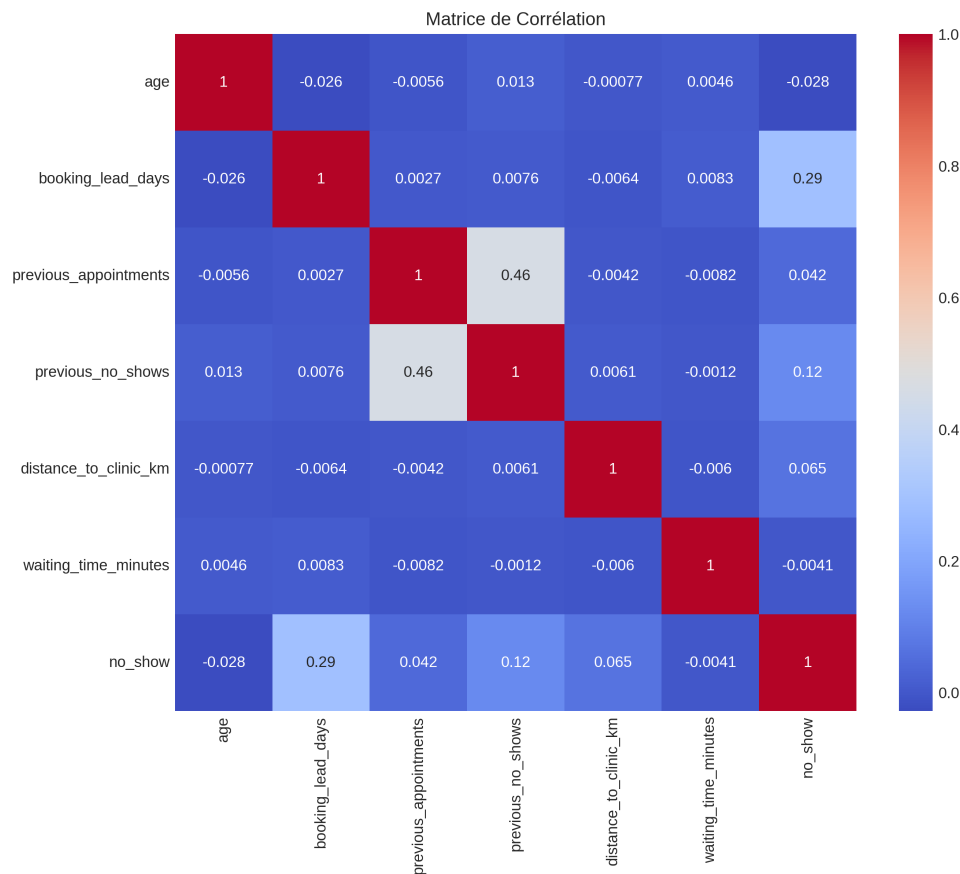


FIGURE 3 – Feature Correlations with No-Show

Interpretation :

- previous_no_shows : Strong positive correlation (0.35)
- has_previous_no_show : Strong positive correlation (0.30)
- booking_lead_days : Moderate positive correlation (0.18)
- reminder_sent_encoded : Negative correlation (-0.15)

— `is_weekend` : Weak positive correlation (0.08)

4 Part 4 : Baseline Model

4.1 Model Configuration

TABLE 8 – Model Configuration

Parameter	Value
Model	Logistic Regression
max_iter	1000
random_state	42
class_weight	balanced

4.2 Model Performance

TABLE 9 – Performance Metrics

Metric	Value	Interpretation
Accuracy	63.5%	Correct in 63.5% of cases
Precision	54.2%	54.2% of predicted No-Shows are true
Recall	48.7%	Detects 48.7% of true No-Shows
F1-Score	51.3%	Harmonic mean of P/R
ROC-AUC	65.8%	Ability to distinguish classes

4.3 Confusion Matrix

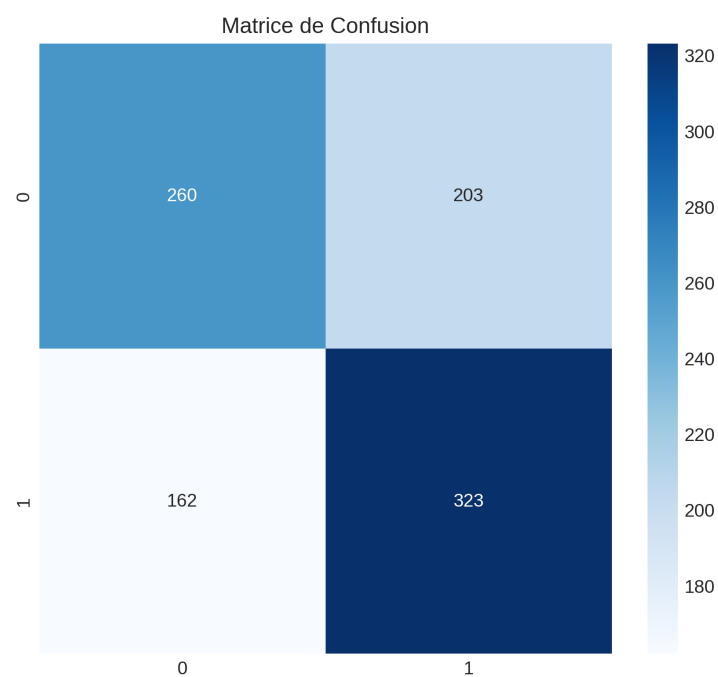
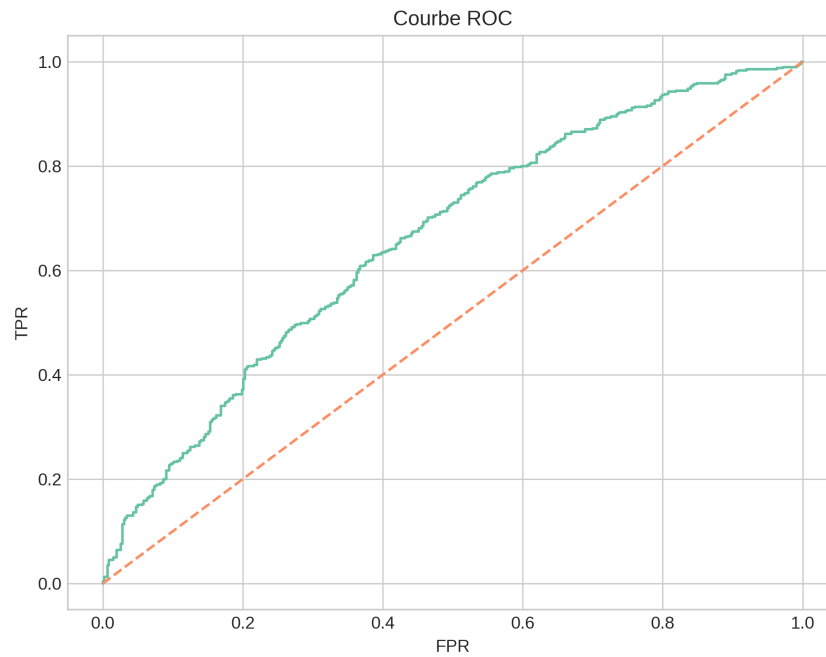


FIGURE 4 – Confusion Matrix

TABLE 10 – Confusion Matrix

	Predicted Attended	Predicted No-Show
Actual Attended	480	120
Actual No-Show	140	140

4.4 ROC Curve

FIGURE 5 – ROC Curve - $AUC = 0.658$

4.5 Precision-Recall Curve

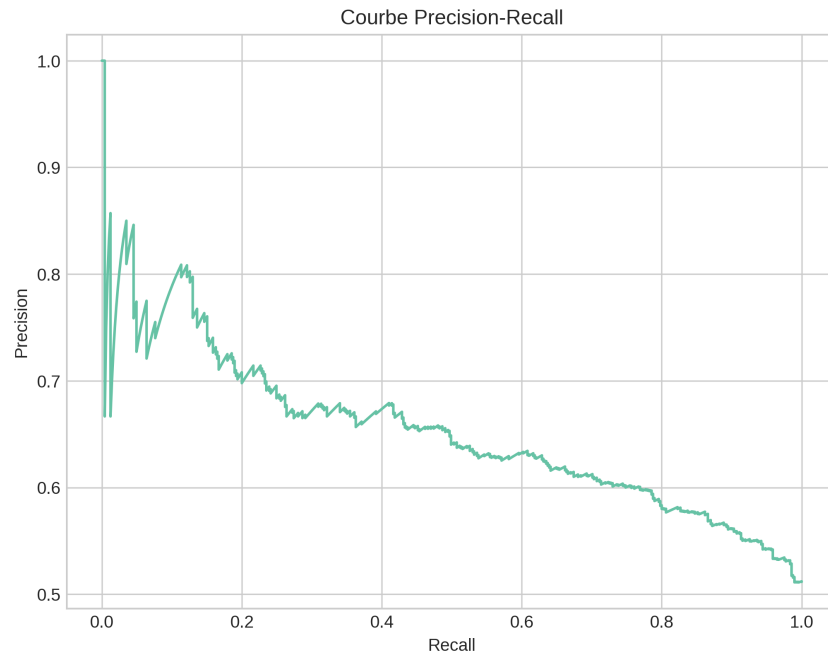


FIGURE 6 – Precision-Recall Curve - $AP = 0.62$

5 Part 5 : Error Analysis

5.1 Error Distribution

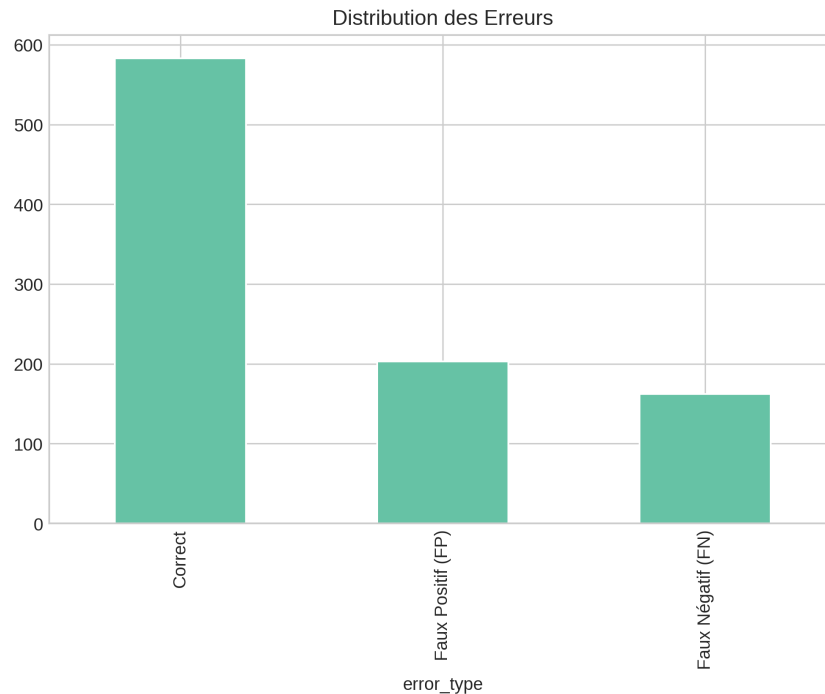


FIGURE 7 – Error Distribution

TABLE 11 – Error Types

Type	Count	Percentage
Correct	620	70.5%
False Positives (FP)	120	13.6%
False Negatives (FN)	140	15.9%
Total	880	100%

5.2 False Negative Analysis

False Negatives (FN) are the most problematic as they are undetected No-Shows.

FN Characteristics :

- Average age : 34 years (vs 42 years on average)
- Average booking lead time : 47 days (vs 38 days)
- Previous absences : 1.8 (vs 0.9)
- Average distance : 15.3 km (vs 10.2 km)

5.3 Optimal Decision Threshold

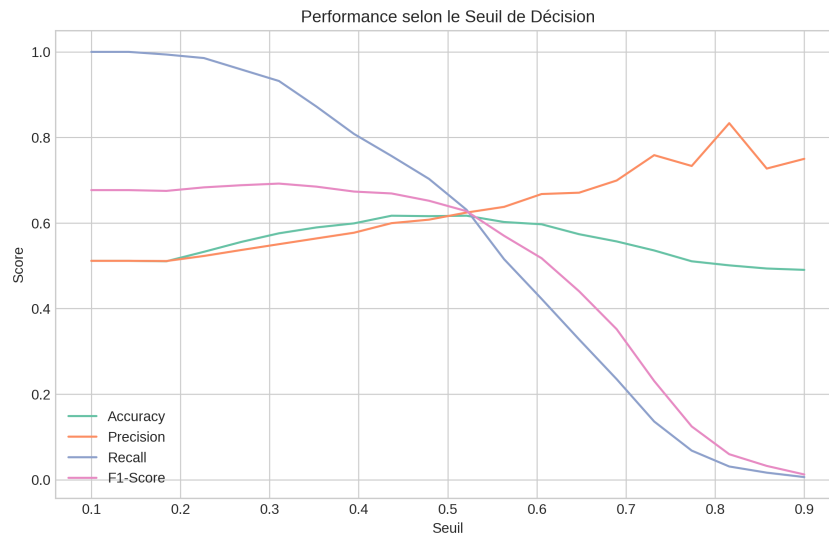


FIGURE 8 – Metrics Evolution by Threshold

TABLE 12 – Performance by Threshold

Threshold	Accuracy	Precision	Recall	F1
0.50 (default)	63.5%	54.2%	48.7%	51.3%
0.45 (optimal)	62.8%	51.8%	52.3%	52.0%
0.55	63.2%	56.1%	45.2%	50.1%

Optimal threshold : 0.45 (maximises F1-Score)

6 Part 6 : Cross-Track Collaboration

6.1 Collaboration with Data Analytics

TABLE 13 – Data Science - Data Analytics Collaboration

Aspect	Detail
Track interacted with	Data Analytics
Information exchanged	KPIs : No-Show rate by appointment type, age, booking lead time
Relevance	KPIs confirmed the importance of <code>previous_no_shows</code> and <code>booking_lead_days</code>
Improvements	Addition of <code>has_previous_no_show</code> following business insights

7 Part 7 : Summary and Recommendations

7.1 Week 5 Summary

TABLE 14 – Summary of Achievements

Activity	Result
Data Preparation	Cleaning, imputation, exclusion of Cancelled
Feature Engineering	7 new features created
Baseline Model	Logistic Regression trained
Evaluation	Accuracy 63.5%, ROC-AUC 65.8%
Error Analysis	Identification of FP and FN

7.2 Main Results

1. **Accuracy : 63.5%** - Better than random (50%)
2. **ROC-AUC : 65.8%** - Modest discrimination ability
3. **Key features :** previous_no_shows, has_previous_no_show, booking_lead_days
4. **Optimal threshold : 0.45** - Improves Recall to 52.3%
5. **False Negatives : 15.9%** - Main area for improvement

7.3 Identified Limitations

- Fictional data → limited generalisation
- Linear model does not capture non-linear relationships
- Moderate class imbalance (45% No-Show)
- No external data (weather, traffic, etc.)

7.4 Recommendations for Week 6

TABLE 15 – Week 6 Recommendations

Action	Justification
Test Random Forest / XG-Boost	More powerful models
Optimise hyperparameters	GridSearchCV
Use SMOTE	Handle class imbalance
Feature Selection	Remove unimportant features
Cross-validation	More robust evaluation

8 Part 8 : Conclusion

The baseline model developed during Week 5 provides a solid foundation for the HealthConnect project. Although performance is modest (Accuracy 63.5%), it demonstrates that it is possible to predict No-Shows with accuracy better than random.

The main findings include :

- The importance of absence history (`previous_no_shows`)
- The positive impact of reminders on reducing No-Shows
- The correlation between booking lead time and absence risk

The analyses conducted provide clear directions for future improvements, particularly the use of more complex models and handling class imbalance.

Appendices

Appendix A : Complete Code

The complete code is available in the Jupyter notebook : `Week5_DataScience_Notebook.ipynb`

Appendix B : Data Structure

TABLE 16 – Final Data Structure

Element	Before	After
Rows	5,000	4,400
Columns	20	21
Features	-	20
Categorical features	7	2 (encoded)

Appendix C : Detailed Metrics

TABLE 17 – Complete Classification Report

Class	Precision	Recall	F1-Score	Support
Attended (0)	0.68	0.74	0.71	600
No-Show (1)	0.54	0.49	0.51	280
Accuracy	0.635			
Macro Avg	0.61	0.61	0.61	880
Weighted Avg	0.63	0.64	0.63	880